Title:	**Hazardous Energy Control Program**		
Document Type:	**Work Instruction**		
Document Number:	**SGOP-0028**	**Revision Number:**	**18**

1.0 PURPOSE

1.1 The purpose of this document is to provide a written procedure to protect associates from unexpected machinery movement/re-energization with a Hazardous Energy Control program (HECP) supported by 'general' and 'equipment-specific' lockout/tagout/tryout (LOTOTO) procedures.

1.2 The program was developed to prevent associates' contact (reach over, under, through or around guards) with dangerous machinery parts (hot, cold, sharp, rotating, ultrasonic; static or dynamic) during intended use and foreseeable abnormal situations.

1.3 This procedure provides the program elements necessary to establish regulatory compliance to the OSHA Standard 29 CFR 1910.147 and the Johnson & Johnson Standard 455 Hazardous Energy/LOTOTO.

2.0 SCOPE

2.1 This policy applies to all facilities presently occupied or to be occupied by Johnson & Johnson Vision Care, Inc. (JJVC) associates in Jacksonville, Florida.

2.2 Any general machinery and equipment operation, equipment service, maintenance, cleaning, or repair where stored energy exists. which could expose associates to unexpected energization, start-up, or release of stored energy or materials shall be covered by this procedure.

2.3 This procedure does not apply to minor servicing/repair activities that are routine, repetitive, and necessary for the normal operation of the equipment if there exists machine or process safeguarding systems under the personal control of the Affected Associate, which are of sufficient integrity to fully protect the associate during the performance of these activities (i.e., protection provided for minor tool changes, adjustments or jam clearing through Zero-Access safety interlocking).

3.0 SUMMARY

3.1 This procedure outlines the requirements to comply with OSHA Standard 1910.147 for the Control of Hazardous Energy, and the Johnson & Johnson Standard 455 Hazardous Energy/LOTOTO. Hazardous Energy Control Procedures (HECPs) is the term we used for JJVC machine specific Lockout/Tag-out procedures. The Hazardous Energy Control Program consists of the following elements:

- Equipment Inventory requirements
- HECP requirements
- Authorized and Affected Associate training requirements
- Designated LOTOTO locks and tags and their approved use
- Guidance for locking and tagging machines out of service other than maintenance purposes
- Application of Group LOTOTO
- Exchange of locks and tags at turnover or shift-change
- Outside Service associates or Contractor requirements
- Annual review of HECPs and Program effectiveness requirements
- Procedure for removal of lock applied by associates absent from the facility
- Procedures for new equipment installations, or where no approved HECP exists

3.2 <u>You are required to read this entire SGOP if:</u>

- You supervise people who work on or near energized equipment.
- You are responsible for management of the HECP Program for your department.
- You are responsible for annual reviews of the HECPs or the HECP Program effectiveness.
- You are a JJVC Contractor Representative (refer to **SGOP-0210**).
- You are Authorized to apply LOTOTO.

4.0 GLOSSARY

TERM	DEFINITION
OSHA Standard 29 CFR 1910.147	A standard issued on September 1, 1989, that established safety requirements for the control of hazardous energy sources by means of lockout/tag-out procedures.
HECP	**Hazardous Energy Control Procedures -** A procedure that defines the requirements for the lockout/tag-out of energy isolating devices. It shall be used to ensure that the machine or piece of equipment is isolated from all potentially hazardous energy and locked out and tagged out before associates perform any servicing or maintenance activities where the unexpected energization, start-up or release of stored energy could cause injury.
LOTOTO	**Lockout Tagout Tryout –** the means by which an isolation device is rendered inoperable in a safe position. After de-energization, when stored energy is released and locks & tags are applied always 'Test' by trying to restart (Try Out) the machine/ equipment/ installation under consideration.
Johnson & Johnson Policy for Tag-out	Johnson & Johnson does not accept tag-out in lieu of lockout.
Authorized Associate	Authorized Associates are those individuals who are authorized by management and trained to apply an isolation technique for specific hazard types prior to performing servicing or maintenance tasks in a safe manner. Only Authorized Associates can apply lockout/tag-out devices in order to perform servicing or maintenance.
Affected Associate	Affected Associates are those individuals who work with equipment that requires lockout/tag-out for servicing and maintenance, or in the area where the proposed work will be performed. Only Authorized Associates can apply lockout/tag-out devices in order to perform servicing or maintenance.
Examples of Hazardous Energy Sources	Electricity, pneumatic (air), hydraulic, elevated-machine members, natural gas, water pressure, chemical, steam, etc.
Minor Repair/Servicing	These are basic service tasks performed by an operator, mechanic, electrician or engineer to maintain the equipment or to troubleshoot/diagnose the equipment to ensure that it runs in optimal operating condition. These tasks are considered normal and routine and do not require the removal, replacement, or adjustment of parts that may expose the associate to potential danger. The associate performing these tasks shall be protected by Zero Access compliant guarding that does not allow the machine to run in an unprotected condition.
Strawman Model	Provides four different definitions of service levels, separated in compliance with regulations which help clarify different service situations based on operational needs of when lockout is required and when it is not. The individual tasks defining the operational situations or repairs of a particular machine are developed into a simplified list or chart to clearly communicate the requirements for a particular machine to the associates who perform the lockout.
Single Source Equipment	Equipment that is powered by only one energy source, i.e., electrical or pneumatic.
Stored Energy	Energy (mechanical, thermal, electrical, gravitational) which can exist in any object with the potential to cause inadvertent movement or injury when released uncontrolled (e.g., compressed springs in pneumatic cylinders, compressed air in any pneumatic system, latent heat in seal plates, spaces with raised mechanical parts, etc.).

5.0 REQUIREMENTS

5.1 Each Organization shall develop and maintain a current inventory of all energized equipment; Equipment-specific Hazardous Energy Control Procedures (HECPs) must be developed, documented and implemented to control potentially hazardous energy when associates are engaged in the maintenance and/or servicing of machinery and equipment; all HECPs shall incorporate the eight LOTOTO steps.

5.2 All Authorized and Affected Associates, and their Supervisors and Managers, who work around or with machinery which has the potential for injury related to machine moving parts shall be trained and retrained every two years.

5.3 Every LOTOTO event shall be conducted in accordance with the eight LOTOTO steps; All hazardous energy sources shall be locked out and tagged out; tagout alone is prohibited.

5.4 Authorized Associates and Contractors shall only use JJVC approved HECP LOTOTO locks and tags; HECP LOTOTO Locks and tags shall not be used for any other purpose.

5.5 Equipment removed from service due for purposes other than repair, maintenance or servicing shall be locked out of service with a different style or color lock and tagged “Out of Service;” HECP LOTOTO approved locks and tags shall not be used.

5.6 All group LOTOTO events shall be accomplished using an approved multiple lock receiving device; each Authorized Associate shall apply and remove only his/her own locks and tags; Group LOTOTO events where there are more than three (3) people using an approved Group Lockout Tagout box shall identify a Group LOTOTO Lead associate.

5.7 Associate changes during a LOTOTO event shall be accomplished by the oncoming Authorized Associate applying locks and tags before the off-going Authorized Associate removes locks and tags. All energy isolation devices shall be capable of being locked in the “safe” position.

5.8 Outside Service or Contractor associates shall be required to comply with JJVC LOTOTO Procedures in accordance with the requirements of this SGOP.

5.9 Each department or organization shall complete an annual multidisciplinary review of the HECPs, and LOTOTO Program effectiveness; including a field observation of the Authorized Associates implementing the LOTOTO procedure, and the Affected Associates impacted, during a representative number of equipment-specific LOTOTO; any findings, CAPAs or improvements identified shall be documented in the approved site CAPA system and reviewed with all Authorized Associates who are reasonably expected to implement the equipment specific LOTOTO procedure during the year.

5.10 Each Authorized Associate shall apply and remove only his or her own lock(s); if an exception is necessary the lock shall only be removed by following the procedures in the PROTOCOL FOR REMOVAL OF LOCK APPLIED BY ASSOCIATES NOT AT FACILITY section of this SGOP.

5.11 All equipment must be locked and tagged in accordance with its approved HECP. If a HECP does not exist a Lockout Tagout (LOTOTO) ePTW Work Permit must be completed (**VCF-3110** is the paper backup should the ePTW system be down and may be used with EH&S approval). For new equipment installation, sections D and E of a HECP must be developed and approved by the project manager and EH&S or EH&S designee or a Lockout Tagout (LOTOTO) ePTW must be completed. Comply fully with the LOTOTO Permit to Work Technical Standard, TV-TEC-164912.

6.0 CURRENT EQUIPMENT INVENTORY

6.1 List all equipment on a departmental equipment inventory.

6.2 The equipment inventory shall include at minimum:

- Organization, Department or area description
- Equipment description\Model #,
- Equipment location,
- Machine Guarding Status,
- Machine Guarding Inspection Form\CMMS,
- HECP Document Number,
- Safety Device/PM Document Number,
- Radiation & Non-Ionizing Radiation Level, and
- Refrigerant Type and Quantity. if applicable.

6.3 Refer to **VCT-0212** for the correct format for the equipment inventory.

7.0 HAZARDOUS ENERGY CONTROL PROCEDURES

7.1 Each machine or piece of energized equipment listed on the Equipment Inventory must have a machine specific HECP to prevent associates' exposure to unexpected movement, energization, start-up, or release of stored energy or materials.

7.1.1 Equipment specific HECPs must be developed, documented and implemented to control potentially hazardous energy when associates are engaged in the maintenance and/or servicing of machinery and equipment.

7.1.2 Refer to **VCT-0213** for formatting, when developing or revising HECPs.

7.2 Adding, revising, or retiring HECPS shall be accomplished in accordance with Document Control procedures.

7.2.1 An Authorized Associate shall be involved in the development or revision of all HECPs

7.2.2 The Authorized Associate involved shall be added to the Release Approval panel of the standard workflow for the HECP.

7.3 All LOTOTO events shall be conducted in accordance with the eight LOTOTO steps:

1. Notify all Affected Associates of shutdown and application of LOTOTO.
2. Steps for conducting machinery or equipment shutdown, including clearing of materials, opening of interlocked doors, or other required preparations.
3. Isolation of relevant machinery or equipment energy sources.
4. Application of locks and tags.
5. Verification of isolation (Verify Zero Energy state).
6. Release of remaining stored energy.
7. Try out to verify and confirm de-energization is effective
8. Return to normal operating condition as described in the LOTOTO procedure.

7.4 Each HECP shall identify all present energy sources requiring LOTOTO, the locations and labeling of the isolation devices; how and when to lockout, and the eight LOTOTO steps.

7.5 All maintenance on machinery or equipment shall be accomplished through the Section D Complete Machine Shutdown Procedures unless there is a specific service level appropriate addendum.

7.5.1 The Strawman Model describes the requirements for specific procedures.

7.5.2 Addenda must specifically address the Service Model Level that applies to the task.

7.5.3 Addenda must include the eight LOTOTO Steps applicable to the specific addendum.

Strawman Model

Service Model Level	Examples
Service Level 1 *Basic operations protected by Category 3 Interlocks;* *Task Hazard Assessment/JSA required.*	- Set-up, jam clearing, cleaning, loading raw material, waste removal. - Protection provided for minor tool changes, adjustments or jam clearing through Zero-Access safety interlocking
Service Level 2 *Minor servicing and repair protected by Category 3 Interlocks;* *HECP Addendum Required.*	- Troubleshooting, lubrication, sensor adjustment, calibration, component replacement that does not require component disassembly, line breaking or breaking wire connections above 50 Volts (plug and cord is acceptable)
LOTOTO Line: Above = "Not required" Below = "Required"	
Service Level 3 *Maintenance and Repair covered by HECP Addendum;* *LOTOTO of applicable energy sources is required.*	- Localized maintenance and repair, component disassembly required, line breaking, breaking of wire connections above 50V procedure covered by a HECP Addendum.
Service Level 4 *Maintenance and Repair requiring full HECP Section D compliance.*	- Preventative maintenance or repair not covered by a HECP Addendum. - Preventative maintenance or repair of multiple zones. - Fixed guard removal and/or repair

7.6 Single-energy source desk top, plug-in or portable equipment with one energy source (i.e. overhead projectors, computer terminals, bench top machines, etc.) may utilize an abbreviated single energy source HECP.

7.6.1 The Department Manager and EH&S must approve the abbreviated procedure.

7.6.2 All multiphase electrical equipment or energized equipment above 240 volts, must have an equipment specific HECP.

7.6.3 Authorized Associate shall complete training for single-source HECPs as described in the TRAINING REQUIREMENTS section of this SGOP.

8.0 LOCKS AND TAGS

8.1 Authorized Associates shall only use JJVC approved HECP LOTOTO locks and tags.

8.2 Each organization shall standardize the locks and tags to be used, by color, shape and size.

8.2.1 JJVC has determined that all locks used for lockout shall be red in color.

8.2.2 All new purchases of locks shall be either Master Locks or American Locks.

8.2.3 Red color locks shall not be used for any other purpose, i.e.: lockers, toolboxes, cabinets, out of service, etc.

8.2.4 The locks and tags shall be durable and difficult to remove.

8.2.5 Each lock shall only have one key that is controlled by the associate using the lock.

8.2.6 Tags should be laminated and specifically designed for use in Lockout/Tag-out applications; at a minimum the tag shall provide space for the person's name and a date. Authorized Associates shall clearly print their name and the date of application of the LOTOTO on the tag.

8.3 Out of service equipment (decommissioned or new installations prior to safety clearance, etc.) shall not be locked out with a red HECP LOTOTO lock and tag. Instead, a different color lock will be used with a tag which states, "Out of Service" and include Point of Contact information (i.e.: name, phone number, email, etc.).

8.4 Each organization shall maintain an adequate supply of locks, tags, and other lockout devices for the equipment and Authorized Associates within their area.

8.5 LOTOTO procedures shall be used when components of a safety system (E-stop, light curtain, door safety switch, contactors, safety relays, safety PLC cards, etc.) are to be added, repaired, replaced or removed (permanently or temporarily) for maintenance or as required by an applicable MOC (**VCF-1816**). Once these change(s) are implemented or repairs/replacement are complete, complete the following prior to operational use:

8.5.1 Prior to restoration of power, guarding for the entire affected machine shall be in place and documented using **VCF-1756**.

8.5.2 After power is restored, all safety devices affected by the change or repair/replacement, shall be operationally tested using the safety device PM for that component(s) or a safety matrix as applicable, documentation requires two (2) signatures.

8.5.3 The results of these tests shall be documented in the approved automated maintenance tracking system, such as CMMS (as an emergency work order) or in VIBES (as a deliverable for a Change Request Assessment).

9.0 OPERATION OF ELECTRICAL DISCONNECTSAND BREAKERS

9.1 Electrical disconnects shall be operated, both off and on, using the Left (Off) Hand Rule in accordance with **SGOP-0022**.

1) Stand to the side of the disconnecting device so you are not in front of the enclosure.
2) Grab the disconnecting device operating handle with your LEFT hand if facing the disconnecting device, RIGHT hand if facing away from the disconnecting device.
3) Turn your body and face away from the disconnecting device.
4) Close your eyes.
5) Take a deep breath and hold it.
6) Move the disconnecting device operating handle to the On or OFF position.

9.2 Operation of electrical disconnect and breakers (both off and on) require the minimum electrical PPE per **VWA-0839**.

9.3 Each organization shall maintain an adequate supply of PPE and is required to train all Authorized Associates in its use.

10.0 GROUP LOCKOUT/TAGOUT

10.1 If more than one individual is required to LOTOTO equipment, each Authorized Associate shall place his or her own personal LOTOTO device(s) on the energy isolating device(s).

10.1.1 When an energy isolating device cannot accept multiple locks and tags, a multiple lockout device (hasp or tree) may be used.

10.1.2 When an Authorized Associate is no longer participating in the work activity requiring the LOTOTO application, that associate shall remove his or her lock own and tag.

10.2 Use of a Group LOTOTO Box is permitted for a job that involves more than three people and the application of individuals applying their LOTOTO to source is not safe or practical; a Group Lockout Box may be implemented if the following conditions are met:

1. Group LOTOTO boxes must have a transparent cover, so all locks and keys are visible.
2. Each lock must be serialized to its corresponding key and have a corresponding LOTOTO Tag.
3. Each tag must identify the specific Group LOTOTO Box where the key is housed.
4. Each key must reference the specific lock to which it operates and must be easily identifiable.
5. An Authorized Associate shall be designated as LOTOTO Leader for the project/area to initiate the LOTOTO as per approved HECP or LOTOTO Permit to Work.
6. Locks and tags may only be applied to the isolation devices by the LOTOTO Leader;
7. The LOTOTO Leader must perform the verification steps from the HECP or LOTOTO Permit and then hang the keys in the Group LOTOTO Box and apply his or her own personal LOTOTO lock and tag to the Group LOTOTO Box.
8. Each isolation device/lockout point identified in, and according to section D (or applicable addendum) of the machine specific HECP or LOTOTO Permit must have a lock and a tag – the tag must identify the lock to the applicable Group LOTOTO Box.
9. Each lock and tag must meet the requirements of the Locks and Tags section of this document.
10. Prior to hanging a lock and tag on a group lockout box, each Authorized Associate must verify there is a lock and tag on each isolation device identified in Section D, or the applicable addendum, of the HECP or the LOTOTO Permit and perform the verification steps.
11. Each Authorized Associate working on the equipment must apply his or her own lock and tag to the Group LOTOTO Box.

12. Each individual must remove his or her own lock and tag from the Group LOTOTO Box; otherwise, follow the PROTOCOL FOR REMOVAL OF LOCK APPLIED BY ASSOCIATE NOT AT FACILITY section of this document.
13. Locks and tags shall only be removed from the equipment by the LOTOTO Leader for the project/area once all other Authorized Associate locks and tags are removed from the Group LOTOTO Box,
14. The LOTOTO Leader is responsible to return the equipment to normal operating conditions in accordance with Section E (or applicable addendum) of the machine specific HECP or the LOTOTO Permit.

11.0 SHIFT CHANGES

Shift changes shall be coordinated by the Authorized Associate designated responsible for the project/area in charge of individual lockout and/or the LOTOTO Leader for a group LOTOTO box. This includes:

- Changing Locks - Incoming Authorized Associates must place their own locks and tags on before outgoing Authorized Associates remove their own locks and tags.
- Reverifying the de-energized state of equipment or machinery being serviced.
- Notification of start-up and testing to be performed, if applicable.
- Changes in the job that affect the LOTOTO procedures (HECP or LOTOTO Permit).

12.0 PROTOCOL FOR REMOVAL OF LOCK APPLIED BY ASSOCIATE NOT AT FACILITY

The following are the steps to take when removing a lock applied by an associate who is not present at the facility:

NOTE: The Department Manager or Supervisor is responsible for supervising this activity. They may utilize other technical support associates, as they deem appropriate.

Step	Action
1.	Ensure every reasonable effort is made to locate or contact the Authorized Associate whose lock(s) and tag(s) were left on the machine.
2.	Visually inspect the entire machine to ensure that no one is working on the machine. In situations where the locked-out energy source isolation is not in direct visual alignment with the equipment, radio communication shall be used to ensure that the equipment is cleared for reenergizing.
3.	Ensure that all machine guards are in place, and that all spare/used parts, tools and materials are removed.
4.	Cut off the lock and tag.
5.	Prior to reenergizing the machine, inspect the machine one more time and if all is clear, state in a clear voice that the machine is about to be reenergized.

13.0 OUTSIDE SERVICE OR CONTRACTOR ASSOCIATES

13.1 Outside service associates or contractors involved in operations relating to equipment or machinery LOTOTO must comply with JJVC hazardous energy control procedures.

13.2 Outside service associates or contractor use of JJVC HECPs shall be coordinated by the JJVC Contractor Representative or Project Manager and documented in the Contractor Safety Packet **VCF-2222** or the Permit to work procedures in accordance with **SGOP-0327** and the LOTOTO Permit to Work Technical Standard, TV-TEC-164912.

13.2.1 The Contractor Company Representative will be trained to be an Authorized Associate by the JJVC Contractor Representative/Authorized Associate and notified as outlined in this SGOP.

13.2.2 The Contractor Company Representative is then responsible to train all contractor company associates to be Authorized Associates for the applicable HECP or LOTOTO Permit.

13.3 JJVC Authorized Associates shall initiate the LOTOTO in accordance with the applicable HECP or LOTOTO permit unless there is an applicable EH&S approved Addendum to the contractor's Safety Packet (**VCF-2222**) or specifically stated on the LOTOTO Permit.

14.0 ANNUAL REVIEW

14.1 A multidisciplinary review of the HECP Program and LOTOTO procedures shall be conducted annually. The review shall include: verification of those designated as authorized, training status and records, lock identification, locks authorized, number of keys/locks, the lock removal process and the out-of-service program. Lessons learned and identified improvements shall be included in the site CAPA system.

14.2 HECP Reviews

14.2.1 To ensure the compliance and effectiveness of the equipment specific HECPs an annual review of each procedure is performed.

14.2.2 This review must be completed with an Authorized Associate at a machine covered by the HECP.

14.2.3 This review must be documented for all equipment specific HECPs within each JJVC department.

14.2.4 The purpose of this review is to ensure that any changes within the last year to the equipment have been accounted for within the HECP.

14.2.5 If any deviances or inadequacies are identified, an entry shall be made in the approved site CAPA system, and the HECP shall be revised and submitted to eDMS within 10 working days to resolve the deficiency.

14.2.6 Document the review of the HECP using the format in **VCT-0214**.

14.3 An annual review of the effectiveness of the HECP Program shall be conducted by each organization utilizing a multidisciplinary LOTOTO Review Team (which must include an Authorized Associate who is trained to implement the LOTOTO procedures).

14.3.1 This will be completed by Department Management with support from EH&S (as required) as part of the annual self-assessment.

14.3.2 The Department Management and EH&S will determine the appropriate sample size for the review; the review must include conducting a field observation of the Authorized Associates implementing the LOTOTO procedure, and Affected Associates impacted, during a representative number of equipment-specific LOTOTO.

14.3.3 The following information shall be documented during this review:

1. The date of the review.
2. Authorized Associates and/or Affected Associates included in the review.
3. The individual(s) performing the review.
4. The identification of the machinery or equipment for which the HECP has been developed.
5. Deficiencies noted. (If deficiencies are noted, management must be notified immediately so that appropriate corrective actions may be implemented.)
6. The individual(s) responsible for the corrective action(s).
7. Projected completion dates for the corrective actions.
8. Deficiencies noted, corrective actions, responsible parties, and projected completion dates shall also be included in the site CAPA system.
9. Refer to **VCT-0215** Lockout Tag-out Demonstration Review of Program Effectiveness Template for format.

14.3.4 Any deficiencies identified shall be reviewed with all Authorized Associates who are reasonably expected to implement the equipment specific LOTOTO procedure during the year.

14.3.5 If associates are identified during the annual reviews that do not have sufficient knowledge, then retraining must be provided.

15.0 EXCEPTIONS TO FULL HECP COMPLIANCE

15.1 JJVC Associates and Contractors are required to use the standardized Lockout/Tagout Permit-to-Work form LOTOTO Permit-to-Work when:

a. no equipment/installation-specific LOTOTO procedure is available, or
b. when the equipment/installation-specific LOTOTO procedure cannot be used, or
c. when contractors who are not trained in local site LOTOTO procedures are involved in activities requiring LOTOTO.

15.2 Refer to **SGOP-0327** Permit to Work.

15.3 Comply fully with the LOTOTO Permit to Work Technical Standard, TV-TEC-164912.

16.0 ASSOCIATE TRAINING AND RETRAINING

16.1 Machine Safety Awareness

16.1.1 All Authorized and Affected Associates who work around or with machinery which has the potential for injury related to machine moving parts, and their supervisors and managers, shall complete the Machine Safety Awareness training prior to commencing work every two years thereafter.

16.1.2 Authorized and Affected Associates are prohibited from working on or around hazardous machines unless they have completed Machine Safety Awareness training within the last 2 years.

16.2 Authorized Associates

16.2.1 Authorized Associates must read all HECP procedures for all equipment that he or she will work on at minimum prior to working with machinery, after HECP update, and every two years thereafter, or as changes are made.

16.2.2 Authorized Associates must also complete machine-specific hands-on training for all equipment that he or she will work on at minimum prior to working with machinery, after HECP update, and every two years thereafter, or as changes are made.

a. The hands-on training shall be conducted by an Authorized Associate assigned by the department manager to perform this function.
b. This training shall be practical, hands-on training which covers exactly how to de-energize, lockout and tag-out equipment as well as how to verify zero energy state for each individual energy source.
c. The hands-on training shall follow the machine-specific HECP.

16.2.3 Hands on training shall be documented using the format in **VCT-0216** or eLMS.

16.2.4 Each organization will maintain a list of associates who are Authorized Associates, and also a separate list of those associates who are Affected Associates.

16.3 Retraining

16.3.1 Retraining will be provided for Authorized and/or Affected Associates as appropriate whenever there is:

- A change in their job assignments
- A change in processes, machines or equipment that presents a new hazard
- A change in the hazardous energy control procedures
- A periodic inspection, assessment or review indicates, or any associate has reason to believe, that an Authorized or Affected Associate has deviated from LOTOTO procedures, due to inadequate knowledge or use of the procedure.

NOTE: All training must be documented.

17.0 RESPONSIBILITES

Title	Responsibility
JJVC Associates	Compliance with all requirements in this document.
Department Manager	1. Ensuring: • Associate compliance to all requirements in this document; disciplinary actions must be taken in accordance with the EH&S Code of Conduct for non-compliance. • Equipment-specific HECPs are developed, documented and implemented to control potentially hazardous energy when associates are engaged in the maintenance and/or servicing of machinery and equipment. • All Authorized and Affected Associates, and their Supervisors and Managers, who work around or with machinery which has the potential for injury related to machine moving parts are trained and retrained every two years. • Every LOTOTO event is conducted in accordance with the eight LOTOTO steps. • Authorized Associates and Contractors use only JJVC approved HECP LOTOTO locks and tags; HECP LOTOTO Locks and tags shall not be used for any other purpose. • Equipment removed from service for purposes other than repair, maintenance or servicing are locked out of service with a different style or color lock and tagged “Out of Service.” • Group LOTOTO events use an approved multiple-lock receiving device, and a LOTOTO Leader is identified for every LOTOTO Event utilizing a Group LOTOTO Box. • Outside service or contractor associates comply with JJVC LOTOTO Procedures in accordance with the requirements of this SGOP. • CAPAs or improvements identified shall be documented in the approved site CAPA system. 2. Maintaining • A current inventory of all energized equipment in the Department. • A list of Authorized and Affected Associates by equipment. • Training records for Authorized and Affected Associates in accordance with the JJVC Records Retention Policy. • HECP program annual reviews and HECP procedure reviews per the Enterprise Record Retention Schedule. 3. Assigning and allocating: • An Authorized Associate to conduct biennial hands-on HECP training for all Authorized Associates using the equipment specific HECPs. • Resources to complete the annual multidisciplinary review • Resources to manage HECP program.
EH&S	1. Assist in hazardous energy control surveys. 2. Designing the Machine Safety Awareness overview training program and making changes to written programs as needed. 3. Completing and documenting completion of the annual multidisciplinary review in the Annual EH&S Self-assessment.